

Rules, Tables, Graphs, and Sensible Domains

Answer Key

Algebra 1

Quick Answers

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|---|--|
| 1. $f(-2) = -11$, $f(0) = -5$, $f(4) = 7$ | 8. $L(0) = 20$, $L(2) = 12$, $L(4) = 4$, time the candle is gone = 5 hr |
| 2. $g(3) = 5$, $g(-1) = -3$ | 9. (a) cost for 9 students = 94, — |
| 3. $x = -4$ | 10. (a) rate of change = 3, (c) the output at input 10 = 32, — |
| 4. $f(0) = 6$, $f(1) = 4$, $f(2) = 2$, x-intercept = 3 | 11. $x = 1, 5$ |
| 5. (a) cost of a 7-mile ride = 17, — | 12. (a) area when one side is 6 = 84 m ² , — |
| 6. (a) $p(6) = 9$, (b) input giving output 0 = $\frac{3}{2}$ | |
| 7. $x = -5$ | |
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What is verified

8 of 12 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 5, 9, 10, 12. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 1

HSF-IF.A – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12

Difficulty 1–5 of 5 across 27 tagged checks.

1. $f(x) = 3x - 5$: complete the table for $x = -2, 0, 4$.

Solution: Substitute each input in parentheses. $3(-2) - 5 = -6 - 5$, so

$$3(0) - 5 = 0 - 5, \text{ so}$$

$$3(4) - 5 = 12 - 5, \text{ so}$$

$f(-2) = -11$

$f(0) = -5$

$f(4) = 7$

2. $g(x) = x^2 - 4$: find $g(3)$ and $g(-1)$.

Solution: $g(3) = 3^2 - 4 = 9 - 4$, so

Wrap the negative input in parentheses: $g(-1) = (-1)^2 - 4 = 1 - 4$, so

$g(3) = 5$

$g(-1) = -3$

3. $f(x) = 2x + 9$: solve $f(x) = 1$.

Solution: The output is given, so this is an equation: $2x + 9 = 1$. Subtract 9 to get $2x = -8$, then divide by 2. Check: $2(-4) + 9 = 1$ ✓

$x = -4$

4. $f(x) = -2x + 6$: complete the table, graph the line, and read the x -intercept.

Solution: $f(0) = -2(0) + 6$, so

$f(0) = 6$

$f(1) = -2(1) + 6 = -2 + 6$, so

$f(1) = 4$

$f(2) = -2(2) + 6 = -4 + 6$, so

$f(2) = 2$

The plotted points $(0, 6)$, $(1, 4)$, $(2, 2)$ lie on a straight line falling two units for every unit right. It meets the x -axis where the output is 0: solve $-2x + 6 = 0$, giving $2x = 6$.

$x = 3$

5. Taxi fare $C(m) = 3 + 2m$ dollars. (a) Find $C(7)$. (b) Why is $m = -4$ outside the domain?

Solution (a): Substitute $m = 7$: $3 + 2(7) = 3 + 14$.

$C(7) = 17$

(b) — **instructor-judged.** A correct answer says a ride cannot have a negative length, so an input below zero produces a number from the rule but describes nothing in the situation; the sensible inputs are the distances that are zero or greater. Full credit for rejecting negative distance and naming the sensible input set; partial credit for rejecting the negative value with no statement of the domain.

6. A linear function p has $p(0) = -3$, $p(2) = 1$, $p(4) = 5$. (a) Find $p(6)$. (b) Find the input whose output is 0.

Solution (a): The input rises by 2 and the output rises by 4 each step, so the rate of change is 2 per unit and the rule is $p(x) = 2x - 3$. Continuing the pattern one more step from $p(4) = 5$ adds 4.

$p(6) = 9$

Solution (b): Solve $2x - 3 = 0$: $2x = 3$.

$x = \frac{3}{2}$

7. $f(x) = 8 - 3x$: solve $f(x) = 23$.

Solution: $8 - 3x = 23$. Subtract 8: $-3x = 15$. Divide by -3 . Check: $8 - 3(-5) = 8 + 15 = 23$ ✓

$x = -5$

8. Candle height $L(t) = 20 - 4t$ cm after t hours. (a) Complete the table and graph it. (b) Solve $L(t) = 0$.

Solution (a): $L(0) = 20 - 4(0)$, so

$L(0) = 20$

$L(2) = 20 - 4(2) = 20 - 8$, so

$L(2) = 12$

$L(4) = 20 - 4(4) = 20 - 16$, so

$L(4) = 4$

The three points $(0, 20)$, $(2, 12)$, $(4, 4)$ fall on a line dropping 4 cm per hour.

Solution (b): The candle is used up when the height is 0: $20 - 4t = 0$, so $4t = 20$.

$t = 5$ hr

9. Trip cost $C(s) = 40 + 6s$ dollars for s students, van seats 15. (a) Find $C(9)$. (b) Which inputs make sense?

Solution (a): Substitute $s = 9$: $40 + 6(9) = 40 + 54$.

$$C(9) = 94$$

(b) — instructor-judged. A correct answer says the input counts students, so it must be a whole number, and the van seats only 15, so the sensible inputs are the whole numbers from 0 through 15. The rule still returns a value for a fractional or oversized input, but that value describes no possible trip. Full credit for both the whole-number requirement and the upper limit.

10. Table for q : $q(1) = 5$, $q(2) = 8$, $q(3) = 11$, $q(4) = 14$. (a) Rate of change. (b) A rule, and why q is linear. (c) $q(10)$.

Solution (a): Using the points $(1, 5)$ and $(3, 11)$: the output rises $11 - 5 = 6$ while the input rises $3 - 1 = 2$, and $6 \div 2$ gives the rate per unit.

$$\text{rate} = 3$$

(b) — instructor-judged. A correct answer gives a rule equivalent to $q(x) = 3x + 2$ and says the table shows a linear function because equal steps of 1 in the input always change the output by the same amount, 3. Full credit for a correct rule plus the constant-difference argument; partial credit for a correct rule with no reason.

Solution (c): $q(10) = 3(10) + 2 = 30 + 2$.

$$q(10) = 32$$

11. $f(x) = x^2 - 6x + 5$: solve $f(x) = 0$.

Solution: Factor: $x^2 - 6x + 5 = (x - 1)(x - 5)$, since $(-1)(-5) = 5$ and $(-1) + (-5) = -6$. A product is zero only when a factor is zero, so $x - 1 = 0$ or $x - 5 = 0$.

$$x = 1 \quad \text{or} \quad x = 5$$

12. Rectangle of perimeter 40 m, area $A(x) = x(20 - x)$ square metres. (a) Find $A(6)$. (b) Why is the domain $0 < x < 20$?

Solution (a): Substitute $x = 6$: $6(20 - 6) = 6 \times 14$.

$$A(6) = 84 \text{ m}^2$$

(b) — instructor-judged. A correct answer says both side lengths must be positive: the input itself must exceed 0, and the other side, $20 - x$, must also exceed 0, which forces the input below 20. Together those give every input strictly between 0 and 20. Full credit for both inequalities and the conclusion; partial credit for rejecting only the negative inputs.